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0.0.1 Agent-based economics

This thesis uses agents to look for spatial economic outcomes from actors with heterogenous locations. There is very little in the way of 'complex' behaviour, and where it does exist (for instance in the interaction of price-setting and production), it is not the focus of analysis. This is not to deny the uses of **Complex Adaptive Systems (CAS)** theory. Evolutionary ideas are particularly important for developing an understanding of how economic growth actually functions (Martin and Sunley 2007, and see section 0.0.4). The evolution of diverse market structures themselves come under the microscope (see e.g. Mirowski 2007a discussing 'marketomata'). But a perceived need to keep **Agent Based Modelling (ABM)** and complexity umbilically linked cuts off some of the more mundane - yet crucial - uses of heterogeneity.

In Otter *et al*'s agent paper presenting an economic geography model, for instance, it is suggested that "regional economics and geography" are apparently "not sufficient to explain the complex spatial patterns, such as clusters and sprawl, that we encounter" (Otter *et al*. 2001 p.1). They argued it is possible to use "complexity as underlying explanatory variable" (ibid). This generally translates into models that claim to be 'validated' if they manifest some element considered complex. (Not a surprising conclusion given some of the theories of explanation that have grown along with **CAS** theory; see section **??** on Epstein.) The 'complexity' cart seems to be pulling the research horse in some cases. Boschma and Martin, for example, ask, "in what sense can complexity theory notions (or metaphors), such as the emergence, self-organisation, criticality and so on, be used to conceptualize the economic landscape?" (Boschma and Martin 2007 p.541)

As O'sullivan and Haklay say, **Agent-based Computational Economics (ACE)** is driven by "a view of the 'economy as an evolving complex system' promoted by the Santa Fe Institute", accompanied by a "widespread disillusion with neoclassical equilibrium economics" O'Sullivan and Haklay 2000 p.1412. Disillusion is a gentle term: the reality for some researchers is a rather more 'year zero' feel, seeing **CAS** as "a pioneering break from a moribund Newtonian worldview" (Manson 2001 p.412). Mirowski, for example, argues (paraphrased by Blaug) that "the whole of neo-classical economics ever since the **marginal revolution** has been an attempt to create an economics that emulates all the essential features of of nineteenth-century physics." (Blaug 1997 p.284)

As Blaug notes, this is rather strong; there was no conscious attempt to emulate physics. Nevertheless, from a **CAS** perspective, the faults with 'traditional' economics are seen to be self-evident. What Conlisk calls the "strange sacrifices required for the 'ritual purity' of optimisation-only models" (Conlisk 1996 p.686) are no longer considered necessary. While understandable at a time when no computers were available, the feeling is - as Mirowski says - economists should "kick the habit of their physics envy and join the 21st century by rethinking the importance of computation and evolution in the way that they approach markets" (Mirowski 2007b p.359). Edmonds goes even further, suggesting that numerical representation itself is questionable. For example, he criticises numerical representation of variety on the basis that not all the dynamics

associated with variety (such as evolutionary dynamics) can be described numerically

need section on GE, and to mention its roots and a siren to call economists to geography...

As section x discussed, **Geographical Economics (GE)** shares this 'Newtonian' lineage. Unsurprisingly, then, there is very little overlap between the research agendas of **GE** and **ACE**. **ACE** as a distinct sub-discipline of **ABM** takes it name from the broader school of computational economics. Despite arguments that **ACE** is perfect for re-examining the Marshallian roots of economics (Leijonhufvud 2006), it has tended to follow the 'Santa Fe' research agenda. One editorial contains a good checklist of the concepts of this agenda: "complexity, evolution, auto-organisation... emergence... bounded rationality, inductive reasoning" (Consiglio 2007 p.vi).

The economic coordination problem in particular lends itself to **ABM**'s default focus on interacting agents. Indeed, Tesfatsion defines **ACE** as "the computational study of economies modelled as dynamic systems of interacting agents". The natural question for **ACE** theorists thinking about markets is then: "who does the job of the so-called market adjustment?" (Posada *et al*. 2007 p.102). This requires the modeller to "analyze explicitly how agents interact with each other" (Kirman and Vriend 2001 p.460) Howitt, for one, sees this requirement as a key strength of the approach: "one of the virtues of the **ACE** approach to economics... is that it forces one to make explicit the mechanisms through which individual actions are coordinated, for better or worse" (Howitt 2006 pp.1068). Agents, it is hoped, force theorists to crack open the problem and look at the dynamics inside.

Hayek's short essay, 'the use of knowledge in society' (Hayek 1945 p.529), is something of a talisman for this way of thinking. In particular, his strong aversion to macro-level assumptions fits **ACE** perfectly. He insisted that "we must show how a solution is produced by the interactions of people each of whom possesses only partial knowledge" (Hayek 1945 p.530) - articulating both an assumption that the 'solution' arises from micro-level interactions, and that it must come from 'boundedly rational' actors. Epstein's 'generativist's question' (discussed in-depth in section **??**) - "how could the decentralised local interactions of heterogeneous autonomous agents generate the given regularity?" (Epstein 2006 p.5) - echoes Hayek's take. Some theorists (Vriend 2002 p.2; Miller and Page 2007) can imagine that, had Hayek only been able to access present technology, he would surely have embraced it, and complexity theory along with it.

This Austrian tilt to **ACE** is further reason for the lack of overlap with the kinds of questions **GE** asks. A good example of how this manifests itself is the criticism levelled at the concept of equilibrium. As Blaug notes, "the desideratum of any economic theory is the delineation of an equilibrium end-state" (Blaug 2009 p.224). These end-states are analysed in the same way that basic physical models of stationary systems assume all forces balance to zero and are then able to deduce force values required for that stationary state. In-keeping with the attacks on its Newtonian underpinning, Colander argues:

> "the self-correcting 'stability' vision cultivated by economic pedagogy is problematic in several respects. First and foremost, it is simply wrong: stability is not the norm in complex systems" (Colander and Rothschild 2010 p.286).

For example, the idea of the 'Walrasian auctioneer' gets set up often as a target. It is described as an actual mechanism: a centralised adjudicator capable of mediating a process of 'groping' towards a set of market-clearing prices, and knowing the moment when they are all correct[1]. Though it does not originate in Walras' own work (Blaug 1997 p.555-6), the auctioneer concept is a good summary of everything **ACE** opposes as implausible. Once such convenient (but, it is argued, impossible) assumptions are shown to be unjustifiable, a Pandora's box is opened:

> "The modeller must now come to grips with challenging issues such as asymmetric information, strategic interaction, expectation formation on the basis of limited information, mutual learning, social norms, transaction costs, externalities, market power, predation, collusion, and the possibility of coordination failure." (Tesfatsion 2006 p.836)

Each of these 'challenging issues' does in fact garner plenty of attention from analytic economics: Stiglitz got his Nobel prize for his work on asymmetric information going back to the seventies (Rothschild and Stiglitz 1976), game theory deals with strategy (and 'predation and collusion' might be considered forms of strategy), and Krugman himself discusses expectation and the role of history as a key element of his geographical work (Krugman 1991). As regards agglomeration economies, the importance of externalities has been discussed in chapter **??**.

ACE models often share one vital feature with the equilibrium methods they reject, however: a market clearing point. This often means that auction-style models are used where markets reach a defined end-point via a bidding process - through many interactions, perhaps, but still terminating at the end of trade. An example would be Kirman and Vriend's fish market model, with an interdependent morning and afternoon's trading (Kirman and Vriend 2001 pp.467) The question of what happens if markets do not 'clear' was asked by Hicks, and Isard understood its implications for trade across space. If all actors are making market decisions on the head of a pin, an auction-style clearing market presents itself as the natural way to think about it. But as soon as actors have heterogenous location, that becomes impossible: 'clearing market' conditions do not hold. As Leijonhufvud puts it, quoting Hicks -

> "In a normal, ongoing market, transactors are not all brought together in a single location and at the same time. Without centralisation and synchronisation, the supply-equals-demand condition 'cannot be used to determine price, in Walras' or

[1] See e.g. Tesfatsion 2006 p.834, where it is argued that "equilibrium values... are determined by market clearing conditions imposed through the Walrasian Auctioneer pricing mechanism; they are not determined by actions of consumers, firms, or any other agency supposed to actually reside within the economy"; (Posada *et al*. 2007.

> Marshall's manner'." (Leijonhufvud 2006 p.1633; see also Ladley and Bullock 2007 p.83-4)

While easier to ignore this problem if geography is not an element of the model, one cannot consider spatially varied trade as 'one market' (Isard 1956 p.43). From an **ACE** point of view, Dibble (2006 p.1516), notes these effects of space on standard market-clearing assumptions. Ladley and Bullock also point out (Ladley and Bullock 2008 p.296) that market actors may be segregated by space, able only to interact with a given subset of others, and that spatial segregation forms natural networks as actors' ranges of interaction overlap. (Indeed, one model does the reverse, and uses actors with randomised location in an R^2 space with a fixed radii as a proxy for building a social network distance model; Hamill and Gilbert 2009, 2010).

It is the 'non-clearing' part of this problem that the thesis model attempts to deal with, rather than the network element. Section **??** already made the argument that transport networks can be replaced with a Euclidean proxy when discussing distance, though only by acknowledging the severe limitations for discussing change this imposes. Ladley and Bullock note Wilhite making a related point: that if network change is slow compared to activity on the network, its structure becomes more important, not less (Ladley and Bullock 2008 p.299 referring to Wilhite 2006). This thesis cannot answer this issue fully as it works within the Euclidean assumptions from section **??**, where it was argued that the success of Euclidean stand-ins is dependent on both the spatial and temporal scale under examination, as well as whether network dynamics are part of the research question. One may be asking, as Epstein does, how "the endogenous connectivity - the topology - of a social network affect its performance as a distributed computational device, one that... computes price equilibria, or converges to (computes) social norms, or converges to spatial settlement patterns such as cities?" (Epstein 2006 p.17-18).

0.0.2 Spatial versus economic agents

A Venn diagram **ACE** and spatial **ABM** would perhaps reveal just as little overlap as between **ACE** and **GE**. Torrens' recent review of '**ABM** in the Spatial Sciences' (2010) illustrates the point. One economic example picked up on is not actually about spatial modelling: Torrens cites Farmer and Foley's Nature piece who argue that the economy is complex, **ABM** can 'do' complexity where equilibrium models cannot, representing something closer to reality; therefore, **ABM** is better:

> "Agent-based models potentially present a way to model the financial economy as a complex system, as Keynes attempted to do, while taking human adaptation and learning into account, as Lucas advocated. Such models allow for the creation of a kind of virtual universe, in which many players can act in complex - and realistic - ways" (Farmer and Foley 2009 p.685-6).

They suggest an effort akin to general climate modelling for the economy is in order, coupling the various social science disciplines in the same way climate models might couple

ocean and atmosphere feedback. Their emphasis on building a 'virtual universe' can be seen in many of the spatial models Torrens surveys (this idea is the subject of section **??**). The cellular automata models surveyed by Torrens in particular have this quality. Crowd and 'swarm' models loom large, leveraging the interactive properties of agents (a point noticed by O'Sullivan and Haklay, 2000 p.1411).

In terms of spatial *economic* models, Torrens cites his own work (among others) - in one example, a technically rich implementation of a combined cellular-automata agent hybrid model (Torrens and Benenson 2005). There appears to be no driving research agenda, however, beyond proving the technical feasibility of doing so, though the ability of disaggregated models to produce 'emergence and self-organisation' is mentioned (p.396).

Elsewhere, much spatial **ABM** is actually targetted as much at 'instrumental' ends, carrying out tasks, as it is 'representational' models in any sense (Brown and Xie 2006 p.941). Rodriguez and Raper (1997) describe the role of spatial agents in instrumental terms: helping users in different ways to optimise use of geographical information systems.

Agent models using explicitly spatial economic ideas that would seem familiar to a **GE** theorist are few. Sasaki and Box use **ABM** to attempt to 'verify' classic spatial analytics like the von Thunen model (Sasaki and Box 2003) by showing how the spatial equilibrium result can come about through individual-level action but, aside from referring to von Thunen himself, it sticks to the **ABM** literature. The work of Christopher Fowler stands out as a theoretically grounded agent model with the explicit aim of examining the **core model**; this is discussed in-depth in section **??**.

0.0.3 Using utility in an agent model

Innovation as evolutionary process for a recent overview see Safarzynska and Bergh 2010 pp.347

The idea of utility is a simple way of thinking through how people react to cost changes, and should be considered as just that. It is often seen to be umbilically linked to what Howitt calls an 'irrational passion for dispassionate rationality', which makes it "easy to dismiss as *ad hoc* or poorly grounded any theory that starts with behavioural rules not explicitly derived from rational foundations" (Howitt 2006 pp.1610) But pinning utility onto rationality is not necessary - any more than the theory of gravity requires planets to love each other.

Section **??** discussed the question of realism of assumptions. Friedman cites an early critic, Thorstein Veblen, pouring derision on the idea of people as "a lightning calculator of pleasures and pains, who oscillates like a homogenous globule of desire of happiness" (Veblen 1898 p.389 quoted in Friedman 1953 p.30). It was argued that pointing out people are simply not like that is not, by itself, a strong enough reason to reject the whole approach.

The **ABM** literature contains opposing views, though the overall bias towards complexity means utility is ignored more often than attacked. Vriend is a rare theorist, both an agent modeller and a careful critic of utility. As he points out, "economic behaviour simply means

that an individual agent chooses (one of) the most advantageous options, given their preferences, in their perceived opportunity set" (Vriend 1994 p.33). Arguments for rational underpinnings are, for Vriend, just "another name for economic behaviour; a question of rhetorics" (Vriend 1996 p.265-6). Feynman's 'levels' argument helps here: this is just working in a framework where something is allowed to be true. As Vriend puts it:

> "Abstracting from an explanation of the individual agent's preferences, and from the mental processes by which he arrives at choices, economics is just a very specific abstraction from reality. Whether these fundamental abstractions are good approximations of reality depends upon the usefulness of the explanatory discourses one can build on it." (Vriend 1996 p.265-6).

going to need some citations about utility from ABMers

This is just reiterating Friedman's point: the theory is essentially tautologous. As Blaug puts it, it does not need a 'hedonistic premise' (Blaug 1997 p.338): there is no need to gain access to people's internal states for utility to be a useful tool for understanding the effect of cost changes. But treating utility theory as a self-consistent tautology is hardly unproblematic. Becker's take on rationality illustrates the issue well:

> "When an apparently profitable opportunity is not exploited, the economic approach does not take refuge in assertions about irrationality Rather it postulates the existence of costs, monetary or psychic, of taking advantage of these opportunities that eliminate their profitability - costs that may not be easily 'seen' by outside observers" (Becker 1976, p.7).

This way of thinking helped Becker produce ideas like rational addiction theory, where the focus becomes precisely about 'rationality as explanation'. This leads, unsurprisingly, to incredulous critiques that argue it "raises the question of how they can be taken seriously" (Rogeberg 2004 p.264). (For a survey, see Melberg and Rogeberg 2010.) Becker's argument also seems to make this approach to utility completely immune to empirical testing. As Conlisk puts it: "whatever the truth about the particular case, economic research often seems to work backwards from empirical findings to whatever utility maximisation will work. Where the empirical arrow falls, there we paint the utility bullseye" (Conlisk 1996 p.685).

Here is a simple thought experiment on utility as a tautology. If a model of allotment production versus growing food in one's backyard were built, how would utility be used? The factors can be reduced to time and distance. Does an actor stand to gain more utility from travelling to the allotment and putting time in there, or by staying at home and growing, thus eliminating travel costs? (This is a similar approach to optimising time used in the models in this thesis.) For a set of actors who can choose between allotment and home-garden plots of similar size, rationally, it seems they should choose to eliminate travel costs and produce in their

own backyard. But what if reality does not conform to this - there is a set of people who choose allotments over their own, perfectly suitable garden. Now one might propose an alteration to the theory, making sure to keep the assumption of rationality: utility is gained from leisure use of the home garden that would be lost by turning it into a vegetable patch. That would need to be outweighed by the disutility of distance before an actor would dig up their begonias.

Thinking about how people react to *space* cost changes actually helps give the discussion some concreteness, and helps avoid the tendency towards fretting over the psychological reality or otherwise of utility. *Ceteris paribus*, nobody wants to travel further than they need to; space costs are not as susceptible to the vagaries of changing taste. More than this, extensive transport economics research is an excellent illustration of the usefulness of the idea in practice. Approaches to utility in transport economics have used the idea of revealed preference

where to define revealed preference?

successfully to say useful things about people's choices. Button points out that "the general conclusion about the idea that some overall budget mechanism governs individual travel decisions, however, must be that, to date, the evidence available still leaves many questions unanswered and the theory is still largely unproved" (Button 2010 p.92). Yet, despite this, many directly practicable ideas have emerged through studying people's reaction to space costs. One that stands out is finding revealed preferences for travel time: "if a person chooses to pay $x to save y minutes then he/she is revealing an implicit value of time equal to at least $(x/y) per minute." This has led to an understanding that -

> "savings in walking and waiting times are valued at between two and three times savings in on-vehicle time - parameters that have proved to be remarkably robust over the years." (**?**)

This is striking: how people value travel time is not simply a function of the time taken. Those willing to commute by car for an hour each way would be much less likely to walk for that length of time. One might counter: utility describes but does not explain this. But again, it is clear that the difference between the theory being a description or an explanation comes down to one's particular purpose, and the level of analysis.

Taking Feynman's quote literally - If a recent piece of research discovered

One last issue is worth mentioning. A particularly troublesome problem is the issue of comparability. Much of the early arguments related to utility revolved around this issue. This included early attempts in experimental economics: even searching for 'utils' and marginal utility by taking milk and bread away from people (Fisher 1927 in Blaug 1997 p.314).

0.0.4 Production

Agent modelling of production faces all the same issues as individual action and utility, not least because often precisely the same functions are used to describe both, and both are required

to maximise those functions given a limited quantity of inputs. The problem of abstracting from the reality of firms' decisions is as thorny as for people. As Blaug notes, utility "no more 'explains' an individual's choices than a production-transformation curve 'explains' the state of technology" (Blaug 1997 p.337). Analysis of production, however, is clearly a more accessible problem than building testable models of people's internal mental processes. From Adam Smith's analysis of pin factories (Smith 1776) to modern work on the collective cognitive processes of productive activities (e.g. Hutchins 1996, a detailed study of a navy navigation crew), the details of production have been a key focus for economics.

The simple approach to production taken in the thesis model needs perhaps even more careful caveats. As Storper notes of the **core model**, attempting to explain agglomeration and growth through " an indeterminate, simultaneous dance of firms, consumer-workers and product varieties and scales... is not very convincing" (Storper 2010 p.317). In particular, linking the quantity of labour input directly to an instantaneous increase in output efficiency clearly does not capture the process of developing those efficiencies. It is important, therefore, to keep claims about modelling that development separate.

Ellerman's essay on Jacobs captures the difference between growth and development nicely: development is not just 'growth' but "differentiation, diversification, and transformation in the products and in the underlying processes of production - all of which might be hidden in the black box of "total factor productivity' ". (Ellerman 2002 p.4). Referring to the common use of capital and labour inputs into production functions (K and L), Ellerman sees development as "more like the process of epigenetic transformation, not blowing up a small balloon - with more K and L - to make a big balloon" (ibid).

This is why reducing development to a difference between Jacobs and Marshall externalities (see section **??**) is dubious: Jacobs' argument for development goes beyond measuring the diversity of sectors in a region, as Glaeser does. This is another case of confusing the levels, however. It is perfectly possible that Glaeser's measuring of diversity is a sensible proxy for just the kind of development Jacobs talks about.

Need to mention various things; some Jacobs; atomistic firms not recognisable from reality; neither people or firms solely atoms interacting in society like atoms in a container. Compare to individualistic critiques; some good stuff from Conlisk: work on evolutionary processes in firms, as well as an overview of research into firms' bounded rationality and "critically shaped by a need to economise on various transaction costs" (Conlisk 1996 p.675).

0.0.5 Coordination between production and utility

Section **??** discussed Fowler's eloquent encapsulation of the co-ordination issues involved. Picking a firm's wage offer as a jumping-in point -

"the amount a firm can offer in wages is dependent on the amount it can produce and the price it receives for its goods. These quantities are affected, in turn, by consumer's wages, which depend on which firm employs them (and whether or not they have found employment

at all.)" (Fowler 2007 p.277)

This tangle of dependencies can be broken down to see how each element in an agent model might relate to each other. Firms can be modelled in ways that avoid many coordination issues, while still helping to shed light on the modelling problem. One approach is to break the cycle of dependencies completely. By providing People with a fixed wage income every turn, issues of money circulation are avoided: firms do not need to set or pay wages, and so payment can appear and disappear again in one day without the need to circulate.

need to x-ref with numeraire stuff / ways to solve money problem / meaning of money etc. Maybe.

The problem of modelling production starts with demand.

In these, a utility function is constrained by a budget. This produces a set of equations giving the 'objective' optimal quantity of goods, given that budget. The approach in **GE** (and many other economic models) is to assume the objective demand implies the correct level of production to meet it, and thus the correct number and scale of firms. In terms of agents, as Leijonhufvud puts it -

> "in this theory utility or profit maximisation is a statement about actual performance, not just motivation... The theory does not leave room for failures to realise the relevant optima." (Leijonhufvud 2006 p.1628)

This is not just an abstract criticism of the constrained optimisation approach: it is a problem that needs solving in agent models using utility. Fowler's approach is to make a tactical choice to allow agents their objective demand. As he puts it, "workers select their optimal bundle of goods without reference to the actual supply of the good" (Fowler 2010 p.10). This avoids the need to solve rationing of limited goods. It also sends "clear signals to firms about the actual level of demand for their goods at the current price" (ibid).

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Todo list

need section on GE, and to mention its roots and a siren to call economists to geography... 2
going to need some citations about utility from ABMers 6
where to define revealed preference? 7
need to x-ref with numeraire stuff / ways to solve money problem / meaning of money etc. Maybe. 9